

Tuan-Anh Le, Ph.D.

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Current position	Research Fellow, Department of Physiology & Biomedical Engineering, Mayo Clinic
Location	Scottsdale, Arizona, United States
Research program	Intelligent Magnetic Robotics - mechanics, magnetic actuation, real-time guidance, sensing, and closed-loop control in dynamic physiological environments
Research metrics	37 published or accepted peer-reviewed journal articles 16 first/co-first author h-index 18 i10-index 29 937 citations ORCID: 0000-0001-5149-626X

EDUCATION

2018-2021	Ph.D., Integrated Technology (Intelligent Robotics Technology Program), GIST, South Korea <i>Dissertation: Development of a Magnetic Particle Imaging-Based Navigation System of Micro/Nano Robots. Advisor: Jungwon Yoon, Ph.D.</i>
2010-2013	M.S., Electrical Engineering, Hanoi University of Science and Technology, Vietnam <i>Thesis: Wind Generator with Vertical-Axis Wind Turbine. Advisor: The Cong Nguyen, Ph.D.</i>
2004-2009	B.S., Electrical and Electronic Equipment Engineering, Hanoi University of Science and Technology, Vietnam <i>Thesis: Design of a Three-Phase Uninterruptible Power Supply. Advisor: The Cong Nguyen, Ph.D.</i>

ACADEMIC AND RESEARCH APPOINTMENTS

11/2022-Present	Department of Physiology & Biomedical Engineering, Mayo Clinic, Scottsdale, Arizona <i>Research Fellow</i>
2021-2022	Gwangju Institute of Science and Technology (GIST), Gwangju, South Korea <i>Postdoctoral Fellow</i>
2018-2021	Gwangju Institute of Science and Technology (GIST), Gwangju, South Korea <i>Research Assistant</i>
2015-2018	Gyeongsang National University, Jinju, South Korea <i>Research Assistant</i>
2014-2015	Gyeongsang National University, Jinju, South Korea <i>Researcher</i>
2009-2014	Thuy Loi University, Hanoi, Vietnam <i>Lecturer</i>

EXTERNAL FUNDING AND GRANT DEVELOPMENT

2026-2027	American Heart Association Postdoctoral Fellowship, Award No. 26POST1565738 - Principal Investigator (Active) <i>AneuFill: Magnetic Field-Assisted Hydrogel Embolic for Device-Free Treatment of Wide-Neck Cerebral Aneurysms.</i> Award amount: \$168,568
2025	NIH/NHLBI K99/R00 Pathway to Independence Award - Principal Investigator/Applicant <i>Magnetically Guided Platforms for Image-Guided Vascular Therapy: Hydrogel Embolics and Hybrid MPI-X-Ray Navigation.</i> Status: Submitted in 2025

RESEARCH PROGRAM

- **Intelligent Magnetic Robotics:** mechanics, magnetic actuation, real-time guidance, sensing, state estimation, and feedback control.
- **Robot-Environment Mechanics:** geometry, compliance, magnetic loading, fluid forces, contact, deformation, deployment/retrieval, and defined operating limits.
- **Guidance and Observation:** fluoroscopy, magnetic particle imaging, magnetic sensing, localization, tracking, and closed-loop intervention.
- **Physiological Validation:** programmable flow models, vascular phantoms, ex vivo human tissue, hemocompatibility, vessel-interaction assessment, and acute/non-survival in vivo testbeds.

SELECTED RESEARCH CONTRIBUTIONS

- **Intelligent Magnetic Robotics.** Developed electromagnetic and permanent-magnet systems for magnetic particles, miniature robots, and therapeutic devices, including focused particle capture, virtual field-free-point manipulation, integrated navigation and heating, robot-arm actuation, and human-scale rotating-field delivery. This work integrates mechanics, field design, sensing, control, and mechatronic implementation.
- **Real-Time Guidance and Magnetic Sensing.** Advanced amplitude-modulation magnetic particle imaging from small-animal to open-sided and human-scale systems using superconducting selection coils. Integrated MPI and

fluoroscopy with localization, tracking, and robotic navigation to support observation, state estimation, and closed-loop operation.

- **Robot-Environment Mechanics and Physiological Validation.** Led EndoBot from compliant mechanical design through magnetic actuation, fluoroscopic guidance, deployment and retrieval, and physiological-flow evaluation in vascular phantoms, ex vivo human umbilical veins, and the rat inferior vena cava, with hemocompatibility, device-integrity, and vessel-interaction assessments. This foundation supports the AHA-funded AneuFill program.

TEACHING AND MENTORING

2026-Present	International doctoral co-supervisor through collaboration with Nazarbayev University Co-supervise a Ph.D. student in magnetic particle imaging.
2023-Present	Research Mentor, Mayo Clinic and collaboration with Arizona State University Mentor undergraduate researchers and a junior research fellow in medical robotics, magnetic actuation, image-guided intervention, experimental design, data analysis, and scientific writing.
2021-2024	International doctoral co-supervisor through collaboration with Nazarbayev University Co-supervised one Ph.D. student to completion. Thesis: Effects of Scanning Trajectory and Parameters on Magnetic Particle Imaging Quality.
2018-2022	Teaching Assistant, GIST, South Korea Courses supported: Mechatronics; Advanced Micro- and Nano-Robotics for Medical Applications.
2009-2014	Lecturer, Thuy Loi University, Vietnam Developed and delivered lectures and laboratories in Electric Machinery, Introduction to Electrical Engineering, Solid-State Electronic Devices, Automatic Control Engineering, Programmable Logic Controllers, power electronics, and drive systems.
2009-2014	Undergraduate Project Supervisor, Thuy Loi University, Vietnam Supervised more than 25 B.S. graduation projects in electrical engineering, controls, and related engineering topics.

HONORS AND AWARDS

2026	Journal Cover Selection - Advanced Functional Materials, "Physiologically Adaptive Soft Millirobot for Atraumatic Endovascular Therapy"
2021	Top 20 Finalist, Vietnam Golden Globe Award for Outstanding Researchers Under 35
2021	Excellent Research Award for Ph.D. Graduation, GIST
2019-2020	Research Assistant Scholarship Award, GIST
2017	Best Paper Award, Gyeongsang National University
2016-2017	Young Research Award, Gyeongsang National University

TECHNICAL EXPERTISE

Robotics and Mechatronics

Magnetic actuation; soft, rigid, and guide-assisted magnetic microrobots; robot-arm manipulation; image-guided navigation; modeling; state estimation; feedback control.

Modeling and Computation

COMSOL Multiphysics; electromagnetic and multiphysics simulation; finite-element modeling; MATLAB/Simulink; reduced-order and experiment-informed modeling.

Imaging and Sensing

Magnetic particle imaging; X-ray fluoroscopy; magnetic sensing and field mapping; localization and tracking; magnetic hyperthermia; quantitative tracer measurement.

Experimental Systems and Validation

LabVIEW; SolidWorks; instrumentation; experimental system integration; physiological flow loops; compliant models; ex vivo human tissue; hemocompatibility; histology; acute/non-survival in vivo testbeds.

JOURNAL ARTICLES AND MANUSCRIPTS

Accepted for publication (1)

1. **Le T.-A.**, Alabay H.H., Anika S., Hoang G.M., Ceylan H. Human-Scale Rotating Magnetic Field Delivery with Permanent Magnets Under Anatomic and Robotic Constraints. Advanced Robotics Research. Accepted for publication, 2026.

Under review (1)

1. Nguyen L.P.*, **Le T.-A.***, Shehu M.A., Shakhin Y., Do T.D., Improving Spatial Resolution of Amplitude Modulation Magnetic Particle Imaging via Optimal Alignment of Drive and Selection-Gradient Fields. Sensors. Under Review, 2026.

In preparation (4)

1. Alabay H.H., **Le T.-A.**, Ceylan H., Topology-Constrained Autonomous Navigation of Untethered Endovascular Microrobots Using Single-Plane Fluoroscopy.
2. Guven M.N., Wang P., **Le T.-A.**, Singh P., Ceylan H., Amorphous Magnetic Soft Robots for Stable Wide-Neck Aneurysm Embolization.

3. Urbina P.A.M., Choi S.R., **Le T.-A.**, Brady J.T., Faubion W.A., Ceylan H., Bose S., Extended Normothermic Perfusion Preserves Viability and Tissue Integrity of Ex Vivo Human Intestine.
4. Wang P., Singh P., Guven M.N., **Le T.-A.**, Ceylan H., An Integrated Rat Neurointerventional Platform for Preclinical Evaluation of Endovascular Devices in Wide-Neck Aneurysm Therapies.

Published peer-reviewed journal articles (36)

1. Wang P.*, **Le T.-A.*** (co-first authors), Singh P., Albadawi H., Alabay H.H., Ceylan H. Non-Survival Rat Endovascular Testbed for Early-Stage Evaluation of Untethered Magnetic Microrobots. *MethodsX*. 2026;16:103906. DOI: 10.1016/j.mex.2026.103906.
2. Nguyen S., **Le T.-A.**, Guven M.N., Lu C., Alabay H.H., Ceylan H., Fluoroscopic Control of a Magnetorobotic Capsule for Precision Gastrointestinal Sampling and Delivery. *iScience*. 2026;29(3):115007. DOI: 10.1016/j.isci.2026.115007.
3. **Le T.-A.**, Alabay H.H., Singh P., Austin M.G., Urbina P.A.M., Misra S., Ceylan H., Physiologically Adaptive Soft Millirobot for Atraumatic Endovascular Therapy. *Advanced Functional Materials*. 2026;36(35):e31400. DOI: 10.1002/adfm.202531400. Selected for the journal cover image.
4. **Le T.-A.**, Tahir M.U., Cao T.L., Yoon J., Development of an Open-Sided Magnetic Particle Imaging System With Amplitude Modulation-Based Reconstruction. *IEEE Transactions on Instrumentation and Measurement*. 2025;74:1-12. DOI: 10.1109/TIM.2025.3568946.
5. Do T.D., Mukhatov A., Tolebay A., **Le T.-A.**, Pham T.T., A Comprehensive Analysis of the Impacts of Image Resolution and Scanning Times on the Quality of MPI-Reconstructed Images. *Scientific Reports*. 2025;15(1):5519. DOI: 10.1038/s41598-025-89296-3.
6. Bui M.P., Park M., **Le T.-A.**, Yoon J., Real-Time 3D MPI-Based Navigation Scheme for Microrobots With Flexible Field Free Point Trajectories and Virtual FFP Intuitive Manipulation. *IEEE Robotics and Automation Letters*. 2024;9(12):10811-10818. DOI: 10.1109/LRA.2024.3474471.
7. Alabay H.H., **Le T.-A.**, Ceylan H., X-Ray Fluoroscopy Guided Localization and Steering of Miniature Robots Using Virtual Reality Enhancement. *Frontiers in Robotics and AI*. 2024;11:1495445. DOI: 10.3389/frobt.2024.1495445.
8. **Le T.-A.**, Bui M.P., Hadadian Y., Gadelmowla K.M., Oh S., Im C., Hahn S., Yoon J., Toward Human-Scale Magnetic Particle Imaging: Development of the First System With Superconductor-Based Selection Coils. *IEEE Transactions on Medical Imaging*. 2024;43(12):4266-4280. DOI: 10.1109/TMI.2024.3419427.
9. Nguyen K.T., Bui M.P., **Le T.-A.**, Kim S.J., Kim H.Y., Yoon J., Park J.O., Kim J., Magnetic Particle Image Scanner Based on Asymmetric Core-Filled Electromagnetic Actuator. *Computers in Biology and Medicine*. 2024;169:107864. DOI: 10.1016/j.combiomed.2023.107864.
10. Mukhatov A., **Le T.-A.**, Do T.D., Pham T.T., Universal Behavior of the Image Resolution for Different Scanning Trajectories. *Applied System Innovation*. 2023;6(6):103. DOI: 10.3390/asi6060103.
11. Park M., Oh S., **Le T.-A.**, Yoon J., Aggregation Volume Estimator-Based Offline Programming Guidance of Magnetic Nanoparticles in the Realistic Rat-Brain Vasculature Model. *Advanced Intelligent Systems*. 2023;5(9):2300128. DOI: 10.1002/aisy.202300128.
12. **Le T.-A.**, Hadadian Y., Yoon J., A Prediction Model for Magnetic Particle Imaging-Based Magnetic Hyperthermia Applied to a Brain Tumor Model. *Computer Methods and Programs in Biomedicine*. 2023;235:107546. DOI: 10.1016/j.cmpb.2023.107546.
13. Mukhatov A., **Le T.-A.**, Pham T.T., Do T.D., A Comprehensive Review on Magnetic Imaging Techniques for Biomedical Applications. *Nano Select*. 2023;4(3):213-230. DOI: 10.1002/nano.202200219.
14. Nguyen K.T., Bui M.P., **Le T.-A.**, Nguyen V.D., Yoon J., Park J.O., Kim J., A Magnetic Particle Imaging in Anisotropic Magnetic Manipulation Platform. *Measurement*. 2023;208:112391. DOI: 10.1016/j.measurement.2022.112391.
15. **Le T.-A.**, Bui M.P., Yoon J., Development of Small-Rabbit-Scale Three-Dimensional Magnetic Particle Imaging System With Amplitude-Modulation-Based Reconstruction. *IEEE Transactions on Industrial Electronics*. 2023;70(3):3167–3177. DOI: 10.1109/TIE.2022.3169715.
16. Park M., **Le T.-A.**, Hadadian Y., Yoon J., Effects of Major Guidance Parameters on Aggregated Magnetic Particles During Magnetic Drug Targeting. *Journal of Magnetism and Magnetic Materials*. 2022;564:170110. DOI: 10.1016/j.jmmm.2022.170110.
17. Park M., **Le T.-A.**, Yoon J., Offline Programming Guidance for Swarm Steering of Micro-/Nano Magnetic Particles in a Dynamic Multichannel Vascular Model. *IEEE Robotics and Automation Letters*. 2022;7(2):3977–3984. DOI: 10.1109/LRA.2022.3148789.
18. Bui M.P.*, **Le T.-A.*** (co-first authors), Yoon J., A Magnetic Particle Imaging-Based Navigation Platform for Magnetic Nanoparticles Using Interactive Manipulation of a Virtual Field Free Point to Ensure Targeted Drug Delivery. *IEEE Transactions on Industrial Electronics*. 2021;68(12):12493-12503. DOI: 10.1109/TIE.2020.3039219.
19. Hoshier A.K., Javan S.D., **Le T.-A.**, Yazdi M.R.H., Yoon J., Studies on Aggregated Nanoparticles Steering During Deep Brain Membrane Crossing. *Nanomaterials*. 2021;11(10):2754. DOI: 10.3390/nano11102754.
20. Cao T.L., **Le T.-A.**, Hadadian Y., Yoon J., Theoretical Analysis for Using Pulsed Heating Power in Magnetic Hyperthermia Therapy of Breast Cancer. *International Journal of Molecular Sciences*. 2021;22(16):8895. DOI: 10.3390/ijms22168895.
21. Darwish M.S.A., Kim H., Bui M.P., **Le T.-A.**, Lee H., Ryu C., Lee J.Y., Yoon J., The Heating Efficiency and Imaging Performance of Magnesium Iron Oxide@Tetramethyl Ammonium Hydroxide Nanoparticles for Biomedical Applications. *Nanomaterials*. 2021;11(5):1096. DOI: 10.3390/nano11051096.
22. **Le T.-A.**, Bui M.P., Yoon J., Electromagnetic Actuation System for Focused Capturing of Magnetic Particles With a Half of Static Saddle Potential Energy Configuration. *IEEE Transactions on Biomedical Engineering*. 2021;68(3):869-880. DOI: 10.1109/TBME.2020.3018266.

23. **Le T.-A.**, Bui M.P., Yoon J., Optimal Design and Implementation of a Novel Two-Dimensional Electromagnetic Navigation System That Allows Focused Heating of Magnetic Nanoparticles. *IEEE/ASME Transactions on Mechatronics*. 2021;26(1):551-562. DOI: 10.1109/TMECH.2020.3041729.
24. Park M., **Le T.-A.**, Eizad A., Yoon J., A Novel Shared Guidance Scheme for Intelligent Haptic Interaction Based Swarm Control of Magnetic Nanoparticles in Blood Vessels. *IEEE Access*. 2020;8:106714–106725. DOI: 10.1109/ACCESS.2020.3000329.
25. Hoshidar A.K., **Le T.-A.**, Valdastrì P., Yoon J., Swarm of Magnetic Nanoparticles Steering in Multi-Bifurcation Vessels Under Fluid Flow. *Journal of Micro-Bio Robotics*. 2020;16(2):137-145. DOI: 10.1007/s12213-020-00127-2.
26. Bui M.P., **Le T.-A.**, Yoon J., Development of Rat-Scale Magnetic Particle Spectroscopy for Functional Magnetic Particle Imaging. *IEEE Magnetics Letters*. 2020;11:1-5. DOI: 10.1109/LMAG.2020.2968407.
27. **Le T.-A.**, Bui M.P., Yoon J., Theoretical Analysis for Wireless Magnetothermal Deep Brain Stimulation Using Commercial Nanoparticles. *International Journal of Molecular Sciences*. 2019;20(12):2873. DOI: 10.3390/ijms20122873.
28. Zhang X.*, **Le T.-A.*** (co-first authors), Hoshidar A.K., Yoon J., A Soft Magnetic Core Can Enhance Navigation Performance of Magnetic Nanoparticles in Targeted Drug Delivery. *IEEE/ASME Transactions on Mechatronics*. 2018;23(4):1573-1584. DOI: 10.1109/TMECH.2018.2843820.
29. **Le T.-A.**, Bui M.P., Yoon J., An Optimal Design of an Electromagnetic Actuator for Targeting Magnetic Micro-/Nano-Carriers in a Desired Region. *IEEE Transactions on Magnetics*. 2018;54(11):1-5. DOI: 10.1109/TMAG.2018.2830774.
30. Hoshidar A.K., **Le T.-A.**, Amin F.U., Kim M.O., Yoon J., A Novel Magnetic Actuation Scheme to Disaggregate Nanoparticles and Enhance Passage Across the Blood-Brain Barrier. *Nanomaterials*. 2018;8(1):3. DOI: 10.3390/nano8010003.
31. Hamdipoor V., Afzal M.R., **Le T.-A.**, Yoon J., Haptic-Based Manipulation Scheme of Magnetic Nanoparticles in a Multi-Branch Blood Vessel for Targeted Drug Delivery. *Micromachines*. 2018;9(1):14. DOI: 10.3390/mi9010014.
32. **Le T.-A.**, Zhang X., Hoshidar A.K., Yoon J., Real-Time Two-Dimensional Magnetic Particle Imaging for Electromagnetic Navigation in Targeted Drug Delivery. *Sensors*. 2017;17(9):2050. DOI: 10.3390/s17092050.
33. **Le T.-A.**, Zhang X., Hoshidar A.K., Kim M.O., Yoon J., Simulation Studies of a Novel Electromagnetic Actuation Scheme for Focusing Magnetic Micro/Nano-Carriers Into a Deep Target Region. *AIP Advances*. 2017;7(5):056724. DOI: 10.1063/1.4977018.
34. Zhang X., **Le T.-A.**, Yoon J., Development of a Real Time Imaging-Based Guidance System of Magnetic Nanoparticles for Targeted Drug Delivery. *Journal of Magnetism and Magnetic Materials*. 2017;427:345–351. DOI: 10.1016/j.jmmm.2016.10.056.
35. Hoshidar A.K., **Le T.-A.**, Amin F.U., Kim M.O., Yoon J., Studies of Aggregated Nanoparticles Steering During Magnetic-Guided Drug Delivery in the Blood Vessels. *Journal of Magnetism and Magnetic Materials*. 2017;427:181-187. DOI: 10.1016/j.jmmm.2016.11.016.
36. **Le T.-A.**, Asl H.J., Do T.D., Kim M.O., Yoon J., Band-Stop Filter Analysis and Design for 1D Magnetic Particle Imaging Hybrid System. *Journal of Nanoscience and Nanotechnology*. 2016;16(8):8492-8495. DOI: 10.1166/jnn.2016.12520.

BOOK CHAPTER

1. Hoshidar A.K., **Le T.-A.**, Amin F.U., Zhang X., Kim M.O., Yoon J., Noninvasive Guidance Scheme of Magnetic Nanoparticles for Drug Delivery in Alzheimer's Disease. In: Thanh N.T.K., editor. *Clinical Applications of Magnetic Nanoparticles: From Fabrication to Clinical Applications*. Boca Raton, FL: CRC Press; 2018. pp. 343–364. DOI: 10.1201/9781315168258-19.

SELECTED CONFERENCE PUBLICATIONS AND PRESENTATIONS

1. **Le T.-A.**, Ceylan H. Human-Scale Rotating Magnetic Field Delivery with Permanent Magnets Under Anatomic and Robotic Constraints. 71st Annual Conference on Magnetism and Magnetic Materials (MMM 2026), Honolulu, Hawaii, United States, November 2–6, 2026. Accepted oral presentation.
2. **Le T.-A.**, Alabay H.H., Ceylan H. Fluoroscopic-Guided Magnetic Soft Millirobot for Atraumatic Endovascular Drug Delivery. Robotics Gordon Research Conference, Ventura, California, United States, January 11–16, 2026. Poster presentation.
3. **Le T.-A.**, Bui M. P., Gadelmowla K. M., Oh S., Yoon J., First Human Brain-Scale Magnetic Particle Imaging System with Superconductor. *International Journal on Magnetic Particle Imaging*. 2023;9(1 Suppl. 1):2303032. DOI:10.18416/ijmpi.2023.2303032. Oral presentation.
4. **Le T.-A.**, Hadadian Y., Kim H., Yoon J., Theoretical Three-Dimensional Predictions for Spatial Focusing of Magnetic Hyperthermia Using the Field-Free Point/Line Concept. 13th International Conference on the Scientific and Clinical Applications of Magnetic Carriers, London, United Kingdom, June 14–17, 2022.
5. **Le T.-A.**, Yoon J., Design of an MPI-Based Magnetic Heating Device for Hyperthermia Cancer Treatment. In: *Proceedings of the 36th ICROS Annual Conference*. Yeosu, Republic of Korea, 2021:117–118. Oral presentation.
6. **Le T.-A.**, Bui M. P., Yoon J., Design of a Rabbit-Scale 3D Magnetic Particle Imaging System with Amplitude Modulation. *International Journal on Magnetic Particle Imaging*. 2020;6(2 Suppl. 1):2009067. DOI:10.18416/IJMPI.2020.2009067. Oral presentation.
7. **Le T.-A.**, Zhang X., Yoon J., Development of a Real-Time Two-Dimensional Navigation System of Magnetic Nanoparticles for Targeted Drug Delivery. 12th International Conference on the Scientific and Clinical Applications of Magnetic Carriers, Copenhagen, Denmark, May 22–26, 2018. Oral presentation.

8. **Le T.-A.***, Zhang X.*, Hoshier A. K.*, Yoon J., An Electromagnetic Navigation System with Real-Time 2D Magnetic Particle Imaging for Targeted Drug Delivery. In: Proceedings of the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). Vancouver, Canada, 2017:1895–1900. DOI:10.1109/IROS.2017.8206007. *Equal contribution.
9. Zhang X., **Le T.-A.**, Yoon J., Development of a Magnetic Nanoparticle Guidance System for Interleaved Actuation and MPI-Based Monitoring. In: Proceedings of the 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). Daejeon, Republic of Korea, 2016:5279–5284. DOI:10.1109/IROS.2016.7759776. Interactive presentation with a one-minute teaser presentation.
10. **Le T.-A.**, Hoshier A. K., Do T. D., Yoon J., A Modified Functionalized Magnetic Field for Nanoparticle Guidance in Magnetic Drug Targeting. In: Proceedings of the 13th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI). Xi'an, China, 2016:493–496. DOI:10.1109/URAI.2016.7734089. Oral presentation.
11. **Le T.-A.**, Do T. D., Noh Y., Mahmood A., Kim M. O., Yoon J., Hardware Implementation of a 1D MPI Hybrid System for Targeted Drug Delivery. In: Proceedings of the 5th International Workshop on Magnetic Particle Imaging (IWMPI). Istanbul, Türkiye, 2015:1. DOI:10.1109/IWMPI.2015.7107010.
12. **Le T.-A.**, Asl H. J., Do T. D., Noh Y., Kim M. O., Yoon J., Band-Stop Filter Implementation for a 1D Magnetic Particle Imaging Hybrid System. 26th International Conference on Molecular Electronics and Devices (IC MED), Republic of Korea, 2015.

PATENTS AND PATENT APPLICATIONS

10 patents and patent applications: 8 issued/registered patents (1 U.S. and 7 Korean) and 2 published pending applications (1 PCT and 1 U.S.).

1. Guiding Microrobots Using Virtual Enhancement. Inventors: Ceylan H., Misra S., Le T.-A., and Alabay H.H. International Application No. PCT/US2025/046186; filed Sep. 12, 2025; published as WO 2026/060264 A1 on Mar. 19, 2026; pending.
2. Apparatus for Nanoparticle Thermal Therapy Based on Open Magnetic Particle Image. Inventors: Yoon J., Le T.-A., and Bui M.P. U.S. Application No. 18/446,152; filed Aug. 8, 2023; published as US 2023/0380712 A1 on Nov. 30, 2023; pending.
3. Magnetic Particle Imaging Device Applicable to Human. Inventors: Yoon J., Le T.-A., and Bui M.P. U.S. Patent No. 12,653,414 B2; filed Aug. 8, 2023; published as US 2024/0215847 A1 on Jul. 4, 2024; issued Jun. 16, 2026.
4. Apparatus and System for Sensing and Controlling Superparamagnetic Nanoparticle. Inventors: Yoon J. and Le T.-A. Korean Application No. 10-2019-0065421; filed Jun. 3, 2019; registered; published Jan. 22, 2021.
5. Nanoparticle Control Apparatus. Inventors: Yoon J. and Le T.-A. Korean Application No. 10-2018-0160743; filed Dec. 13, 2018; registered; published Jun. 10, 2022.
6. Magnetic Particle Spectroscopy. Inventors: Yoon J., Bui M.P., and Le T.-A. Korean Application No. 10-2020-0140449; filed Oct. 27, 2020; registered; published Aug. 18, 2022.
7. Open Magnetic Particle Imaging Apparatus. Inventors: Yoon J., Le T.-A., Bui M.P., and Gupta M. Korean Application No. 10-2020-0143808; filed Oct. 30, 2020; registered Mar. 6, 2023; published Mar. 9, 2023.
8. Apparatus for Controlling Fine Particles Using Magnetic Field Force. Inventors: Yoon J. and Le T.-A. Korean Application No. 10-2020-0159654; filed Nov. 25, 2020; registered Jun. 27, 2023; published Jun. 30, 2023.
9. Apparatus for Controlling Magnetic Particles. Inventors: Yoon J. and Le T.-A. Korean Application No. 10-2020-0159521; filed Nov. 25, 2020; registered Jul. 11, 2023; published Jul. 14, 2023.
10. Portable Magnetic Particle Imaging Device. Inventors: Yoon J., Choi S.H., and Le T.-A. Korean Application No. 10-2021-0190350; filed Dec. 28, 2021; registered Oct. 6, 2023; published Oct. 12, 2023.

ACADEMIC SERVICE AND PROFESSIONAL ACTIVITIES

- Associate Editor, IEEE Access (June 2025-Present).
- Guest Editor, Micromachines, Special Issue: Integrated Robotics for Micromanipulation, Diagnosis and Microsurgery (2022-2023).
- Completed more than 60 peer reviews for journals in robotics, mechatronics, medical imaging, biomedical engineering, and advanced materials, including IEEE/ASME Transactions on Mechatronics, IEEE Robotics and Automation Letters, IEEE Transactions on Biomedical Engineering, Advanced Materials, Advanced Science, Advanced Healthcare Materials, Small, and Advanced Intelligent Systems.
- American Heart Association, Member (2023-Present).